

MODULE 05

Test Automation with Tosca

Day 2 · Optional MVP Module



ADOBE STOCK PLACEHOLDER

Suggested search: "automated software testing dashboard (wide hero banner)"

LEARNING OBJECTIVES

- Understand model-based testing for API and UI validation.
- Organize execution lists and quality gates.
- Publish automated test results into delivery workflows.



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Suggested search: "QA tester reviewing test results (supporting image)"

WHY THIS MODULE MATTERS

This module can be delivered as an optional hands-on track once Tosca provisioning is complete.

Otherwise it runs as an instructor-led demo in MVP phase one.

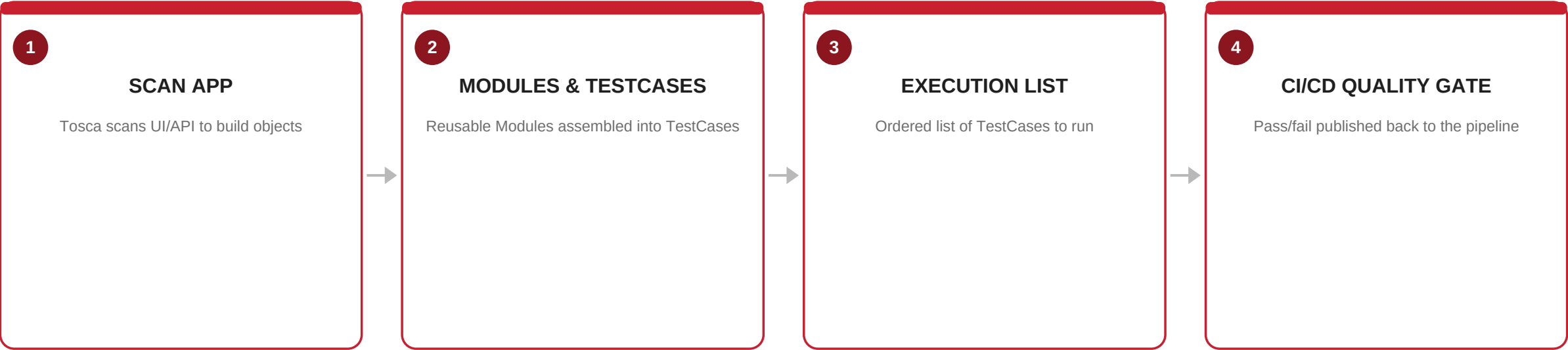


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Suggested search: "software test automation concept (concept image)"

CONCEPT MAP

How a Tosca model-based test run actually moves from the app to a pipeline gate.



FROM THE DOCS: WHAT MODEL-BASED TESTING ACTUALLY MEANS

- Tosca scans an app's UI/API to build a codeless, business-readable automation model — not hand-written scripts.
- Tricentis's own framing: traditional scripted tools are technical and brittle; MBT decouples test logic from the model.
- Modules capture reusable app objects; updating one Module auto-syncs the change across every linked TestCase.
- TestCases are assembled from TestSteps (actions) and TestStepValues (data/verification points).
- Tosca Cloud connects to CI/CD tools like Jenkins via API to automate execution end to end.



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Suggested search: "quality assurance checklist on screen (detail image)"

COURSEWARE FOCUS

- Model-based testing
- Execution lists
- CI integration



ADOBE STOCK PLACEHOLDER

Suggested search: "automated software testing dashboard (detail image)"

TOOLS & RESOURCES

Official Docs

Tricentis Tosca Documentation: Model-Based Testing and ADO Integration

Theoretical basis for MBT (Modules, TestCase Design, Execution Lists).

CI/CD Plugin

Tricentis Tosca Execution Task for Azure DevOps / GitHub Actions

Integrating automated UI/API test runs directly into pipelines.

Target App

Modernized REST API built in Module 4

Live endpoint target for scanning UI and executing API test suites.

SUPPORTING ASSETS

- Target app from Module 4
- Tosca runtime and integration connectors



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Suggested search: "QA tester reviewing test results (reference screenshot)"

LAB 05: TEST AUTOMATION QUALITY GATES



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Suggested search: "software test automation concept (lab / hands-on photo)"

LAB STEPS

Goal: Build and run automation suites and define release quality gates.

Tools: Tosca (installed and licensed) · Target app is the Lab 04 API.

- Step 1 — Scan the target UI and API endpoints.
- Step 2 — Build smoke and regression suites.
- Step 3 — Execute test lists and capture results.
- Step 4 — Publish results to the pipeline or work tracking.
- Step 5 — Define pass/fail quality gate thresholds.
- Validation: the automated suite executes successfully, and gate criteria are documented.



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Suggested search: "quality assurance checklist on screen (hands-on lab photo)"

KNOWLEDGE CHECK: Test Automation with Tosca

What does model-based testing primarily rely on to design test cases?

- A. Random fuzzing of inputs
- B. A model or representation of the application, rather than hand-scripted steps
- C. Manual exploratory testing only
- D. Production incident reports



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KNOWLEDGE CHECK: Test Automation with Tosca

In Tosca's model, what happens when you update one Module used across many TestCases?

- A. Nothing — each TestCase must be edited by hand
- B. The change automatically syncs to every linked TestCase
- C. All TestCases using it are deleted
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SOURCES & FURTHER READING

EXTERNAL DOCUMENTATION

■ Tosca — Model-Based Test Automation

<https://www.tricentis.com/products/automate-continuous-testing-tosca/model-based-test-automation>

REFERENCED IN THIS REPOSITORY

■ `course/mvp-delivery/labs/lab-05-test-automation-quality-gates.md`

RECAP & SUCCESS CRITERIA

- Automated runs execute and publish actionable pass/fail results.
- Tier: Optional MVP Module



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Suggested search: "software test automation concept (team success photo)"